Program : **Diploma in Manufacturing Technology**	
Course Code :**4702**	Course Title: **CNC Programming**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester:**60**

CourseObjectives:

- To familiarize with the basics of CNC systems
- To familiarize with structure of programming and programming codes
- To provide knowledge in different machining cycles in CNC programming

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic science		Mathematics I&II	1,2
Basic science		Applied Physics I&II	1,2
Engineering Science		Engineering Graphics	1
Program core course		Manufacturing Process and Machine tools	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Express the basics of NCSystem, Components, Need and Advantages of CNC Machines.	11	Understanding
CO2	Understand data processing in CNC,Distinguish various Axis motion and Direction Machine controller, Machine Zero, Work Zero and Tool Reference points	12	Understanding

CO3	Prepare Part Programming in lathe using M Codes and G Codes, basic machining cycles and for Turning Jobs	19	Applying
CO4	Prepare part programming in milling using M Codes and G codes, Understanding Drilling cycles and mirroring in CNC milling.	16	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2	2					
CO3	2	3					
CO4	2	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Express the basics of NC System, Components, Need and Advantages of CNC Machines.**		
M 1.01	Describe Numerical control systems	2	Understanding
M 1.02	Explain components of CNC ,Advantages of CNC	3	Understanding
M1.03	Distinguish between conventional and CNC machining	3	Understanding
M1.04	Identify the Need and Application of CNC	3	Understanding
Contents: NC System-definition-components-punched tape-Types of NC System-open loop and closed loop control systems-advantages and disadvantages-CNC-Advantages of CNC-Applications of CNC-Need for CNC-Components of CNC system- feedback system-Open loop and closed loop system-Differences between conventional and CNC machines.			

CO2	**Understand data processing in CNC; Distinguish various Axis motion and Direction Machine controller, Machine Zero, Work Zero and Tool Reference points.**		
M2.01	Understand data processing in system	3	Applying
M2.02	Explain different Axis movements	3	Applying
M2.03	Understand and explain about Program zero`	2	Understanding
M2.04	Understand and explain position Work Zero	2	Understanding
M2.05	Understand and explain tool reference points	2	Understanding
	Series Test – I	1	
Contents: Data processing in CNC system-Data reading-conversion of data-binary, decimal-Axes designation X Y Z axes, directions of axis movement-Rotary axis ABC -right hand thumb rule-co-ordinate systems, position, planes, Cartesian system and polar co-ordinates-incremental and absolute system of programming-Reference Point-Program zero, machine zero, and tool reference point-setting the work part zero point on lathe.			
CO3	**Prepare Part Programming in lathe using M Codes and G Codes, basic machining cycles and for Turning Jobs**		
M3.01	Study M Code and G Code and its application to generate Part programming in CNC lathe	3	Applying
M3.02	Identify and understand program format	4	Applying
M3.03	Prepare simple part programmes in CNC lathe using G code and M codes	4	Applying
M3.04	Understand definition and purpose of machining cycle.	4	Understanding
M3.05	Prepare simple part programmes using cycles- G71,G70,G75,G74,G 76 in turning operations	4	Applying
Contents: Explanation of M codes-G codes-program format-address characters, word-program blocks-numbers-simple part programming in lathe-machining cycles-Definition-purpose-format-G70, G71, G74, G75, G76-Simple programmes.			
CO4	**Prepare part programming in milling using M Codes and G codes, Understanding Drilling cycles and mirroring in CNC milling.**		
M4.01	Study M Code and G Code and its application to generate Part programming in CNC milling	4	Applying
M4.02	Understand about Machine axes in CNC milling	4	Applying

M4.03	Prepare simple part programmes in CNC milling using G code and M codes	4	Applying
M4.04	Describe drilling cycles and understand mirroring operation in CNC milling.	4	Applying
	Series Test – II	1	
Contents: Explanation of G codes and M codes-prepare simple part programmes in CNC milling-Machine axes-4 axis-5 axis-6 axis-Drilling cycles-G73, G81, G82, G83-simple programmes-Canned cycle-definition only-Mirroring-definition and concept.			

Text / Reference

T/R	Book Title/Author
T1	CNC Programming - Dr.S K Sinha-Galgotia Publication New Delhi
R1	Computer Numerical Control Machine – Radhakrishnan P - New Central Book Agency
R2	Computer Numerical Control Machine Tools – Thyer GE – BH Newberg
R3	CNC Technology and Programming- Krar.S -HcGraw Hill

Online Resources

Sl.No	Website Link
1	https://www.cnccookbook.com/g-code-m-code-reference-list-cnc-mills/
2	https://nptel.ac.in/content/storage2/courses/112103174/pdf/mod7.pdf